Analytical Cume
September 14, 2001

This cume tests your ability to propagate uncertainty in laboratory procedures

For use in an iodine titration, you prepare an solution using 0.2222 (±0.0002) g of KIO_3 (214.0010 (±0.0009) g/mole) in 50.00 (±0.05) mL.

1. Find the molarity and its uncertainty with an appropriate number of significant figures.
2. Would the answer be affected significantly if the reagent were only 99.9% pure?

There are two methods by which you could prepare a dilute silver nitrate solution.
Method 1: Weigh out 0.0463 (±0.0003) grams of $AgNO_3$, and dissolve it in a 100 mL volumetric flask.
Method 2: Weigh out 0.4630 (±0.0003) grams of $AgNO_3$, and dissolve it in a 100 mL volumetric flask. Then pipet 10 mL of this solution into a fresh 100 mL volumetric and dilute to the mark. In both methods assume that the uncertainty in the pipets and volumetric flasks is zero, and the only uncertainty is in the balance.

3. Which method produces solution with more precise molarity? Be sure to give a quantitative result. You may take the molar mass of silver nitrate to be exactly 169.873 g/mol.

Consider the titration of 50.00 (±0.05) mL of a mixture containing the ions I^- and SCN^- with a 0.0683 (±0.0001) M silver ion solution.

Which ion will precipitate first in the titration given the solubility product constants are:

$Ksp(AgSCN) = 1.1 \times 10^{-12}$

$Ksp(AgI) = 8.3 \times 10^{-17}$

The first equivalent point is observed at 12.6(±0.4) mL, and the second occurs at 27.7(±0.3) mL.

4. Find the molarity and the uncertainty in the molarity of iodide in the beaker. (You may ignore any error introduced by the solubility products not being zero).
5. Find the molarity and the uncertainty in the molarity of thiocyanate in the beaker. (You may ignore any error introduced by the solubility products not being zero).
6. Why does one molarity have more uncertainty than the other?

Equations you may find useful:

$$e_4^2 = e_1^2 + e_2^2 + e_3^2$$

$$\%e_4^2 = \%e_1^2 + \%e_2^2 + \%e_3^2$$